

Power BI Fundamentals

Power BI



SoftUni Team

Technical Trainers



SoftUni



Software University

<https://about.softuni.bg/>

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The Goal of Visualization

Nobody reads SQL code. Managers want answers.



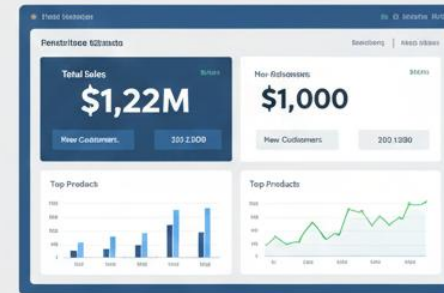
THE 5 SECOND RULE IN POWER BI DESIGN

BEFORE



Problem: Information Overload, Slow Comprehension (More than 5 seconds)

AFTER

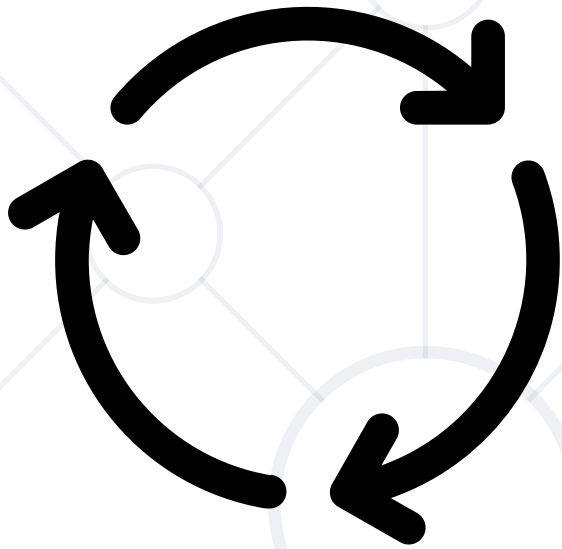


KEY INSIGHT: Sales Increased 20% This Month

Solution: Simplify:
Simplify, Focus on Key Metrics,
Get Insight in < 5 seconds

The 5-Second Rule (Visualized)

- **Before:** A massive Excel spreadsheet with thousands of rows.
- **After:** A clean dashboard with 3 KPIs (Revenue, Profit, Orders) and one Trend Line.
- **Key Takeaway:** We are not just displaying data; we are designing attention.



1. **Get Data** (Connect to SQL/Excel).
2. **Transform** (Power Query).
3. **Model** (Relationships).
4. **Visualize** (Report Canvas). <-- Focus of Today
5. **Publish** (Power BI Service).

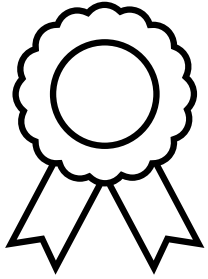
Core Concept

- **Instant Understanding:** A user opening the report should immediately see the most critical information and understand the primary message.
- **Focus on Action:** It's about enabling quick decisions and identifying trends (good or bad) instantly, not deep analysis.
- **Simplicity is Key:** If it takes longer than 5 seconds to get the point, the visualization is too complex and needs simplification.

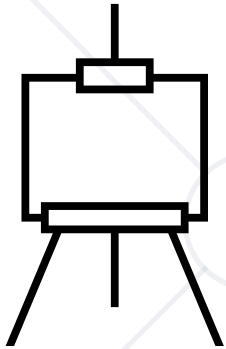


The Three Views (Sidebar)

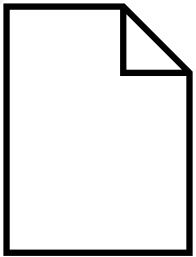
- 1. Report View:** The Canvas. This is where you build visuals.
- 2. Data View:** Spreadsheet look. Use this to inspect rows, sort data, and verify types.
- 3. Model View:** Relationship diagram. We used this in the previous lecture to build the Star Schema.



- The Ribbon (Top): Standard Microsoft tabs (Home, Insert, Modeling).



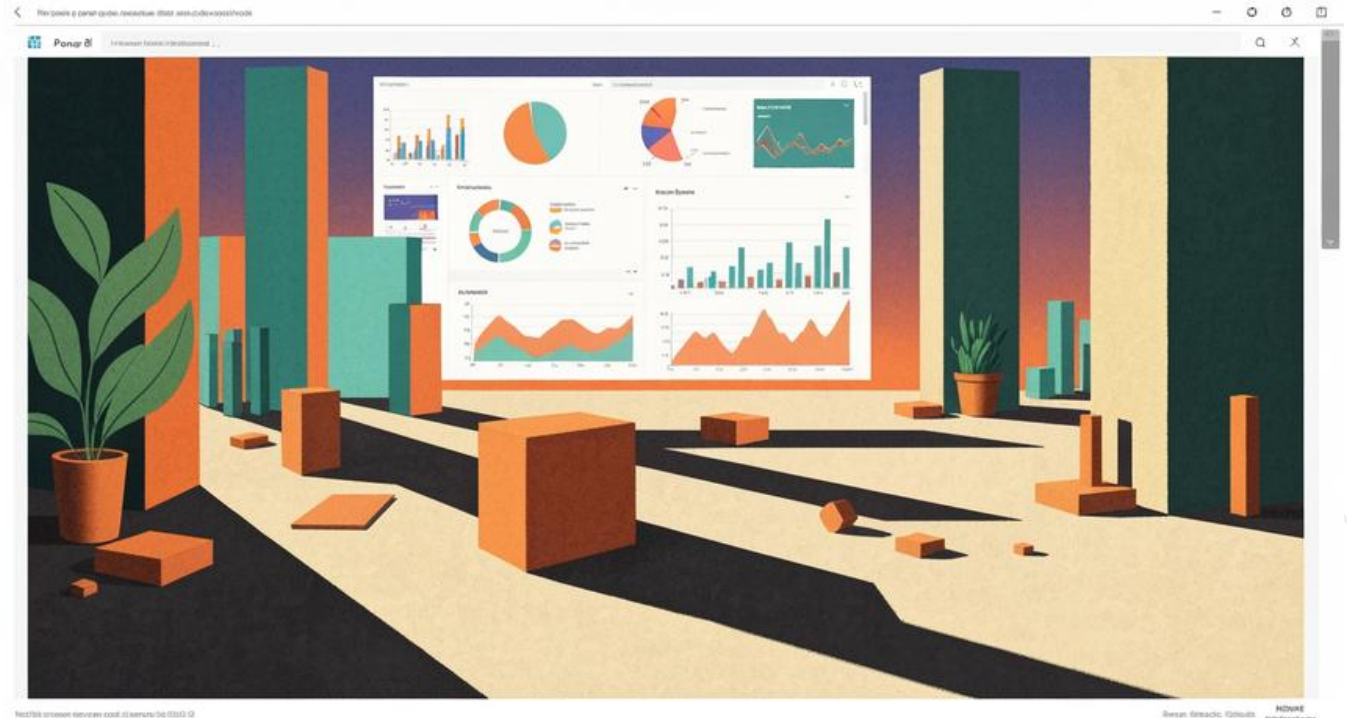
- The Canvas (Center): Your whiteboard. This is what the user sees.



- Pages (Bottom): Just like Excel tabs. You can have multiple dashboard pages.

The Visualizations Pane (Right Side)

- Build Visuals: The icon gallery (Bar, Line, Map, etc.).
- Data Well: Where you drag columns (X-Axis, Y-Axis, Legend).
- Format Visual (Paintbrush): Colors, Fonts, Titles, Borders.



The Data Pane

Displays your tables (Sales, Employees, Customers).

Icons matter:

Σ (Sigma): Numeric field (can be summed).

 (Calendar): Date field (has hierarchy).

 (Globe): Geographic field.



Source: We are connected to Tech_DB (PostgreSQL).

Mode: Import Mode.

- Power BI takes a Snapshot of the data.
- If the database updates, you must click Refresh to see new numbers.



Reminder from Lecture 2:

- 📷 Import = Photo of data (fast, static)
- 🔗 DirectQuery = Live camera (slow, real-time)

Keyboard shortcut: Home → Refresh (or press F5)

Why it matters: Power BI needs to know what the data represents.

`vw_dim_customers[City]` → Data Category: City

`vw_dim_customers[Country]` → Data Category: Country

Data Categories:

- Select City column -> Set Data Category to City.
- Select Country -> Set Data Category to Country.

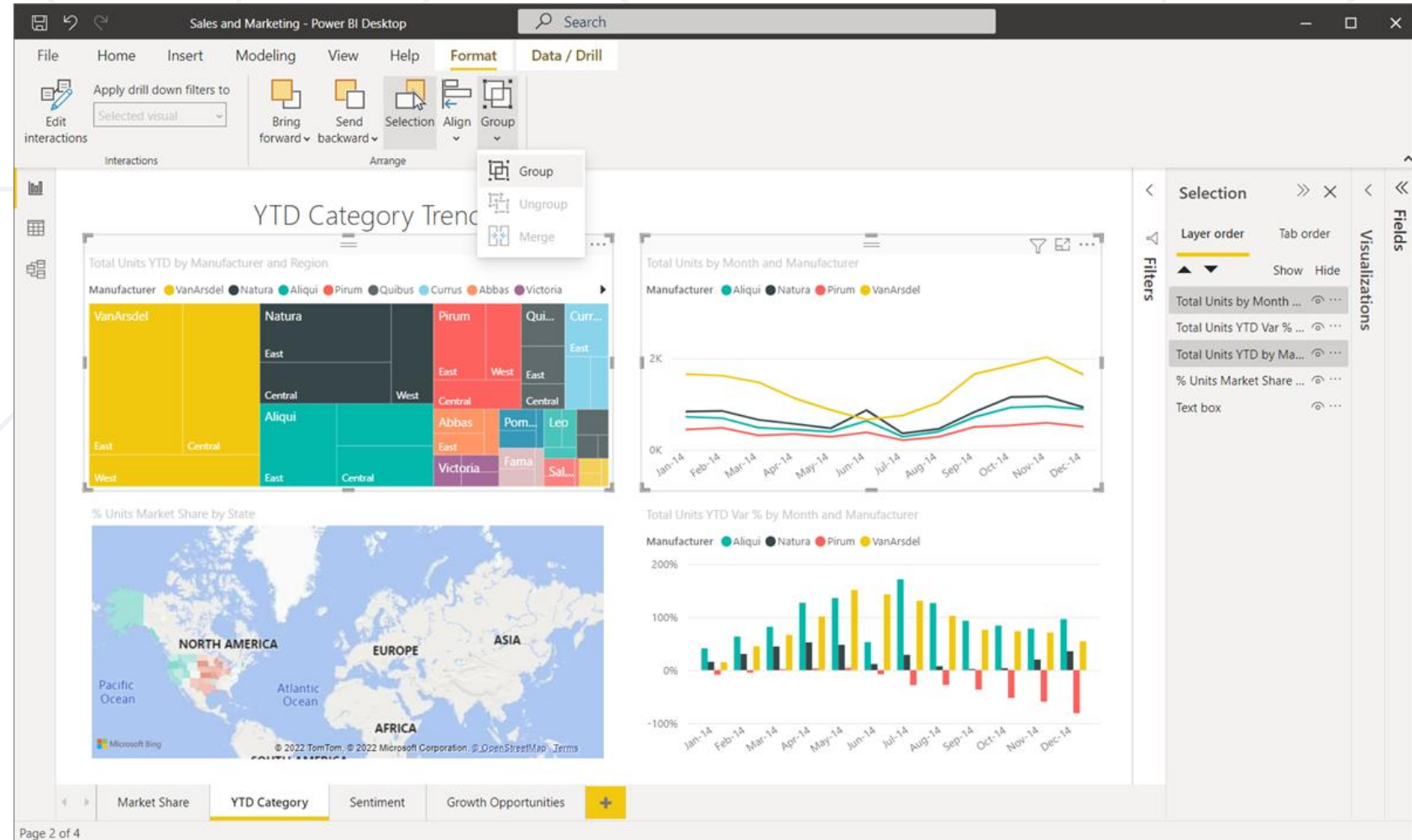
Formatting: Currency, Percentages, Decimals.

Live Demo



Basic Visuals

- Cards
- Bar/Column Charts
- Line Charts
- Maps



Visual is The Headline



No one reads the small text
before checking the headline
to see if the news is good or bad.



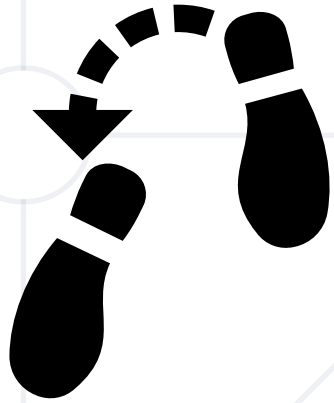
- **What is a Card?** A visual designed to display a single, critical number.
- **Why use it?** For Key Performance Indicators (KPIs) that require immediate attention.
- **The 5-Second Rule:**
 - Place Cards in the **top-left corner** (Z-Layout).
 - Use them for "Total Revenue", "Total Profit", or "Total Orders".
- **Best Practice:** Always format units (e.g., display **1.2M** instead of **1,234,567.89**).

Live Demo



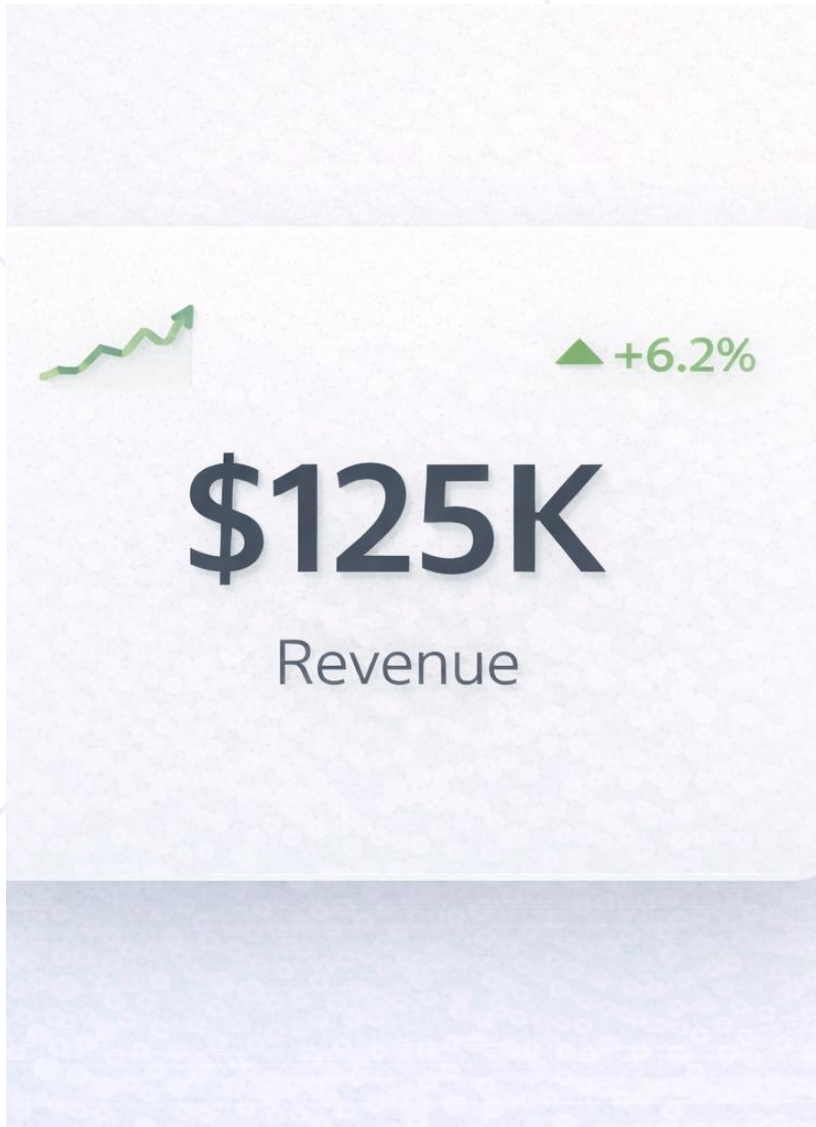
Visualizing Revenue

The CEO wants to see the total turnover immediately.



1. Select the Card (123) visual.
2. From the **Sales** table, drag the **Amount** field to the Fields well.
3. Power BI automatically sums the values (**Sum of Amount**).

Crucial Step - Renaming:



- Double-click the field name in the "Visualizations Pane".
- Rename from "Sum of Amount" to **"Total Revenue"**.
- Result: A clean, professional KPI card.

The KPI Card (123)

- Use Case: The most important single number (Revenue, Total Staff).
- Placement: Top-Left (First thing the eye sees).
- Tip: Always rename the field (e.g., "Sum of Amount" -> "Total Revenue").



Before: "Sum of Amount" (Technical, auto-generated)

After: "Total Revenue" (Business-friendly, clear)

How:

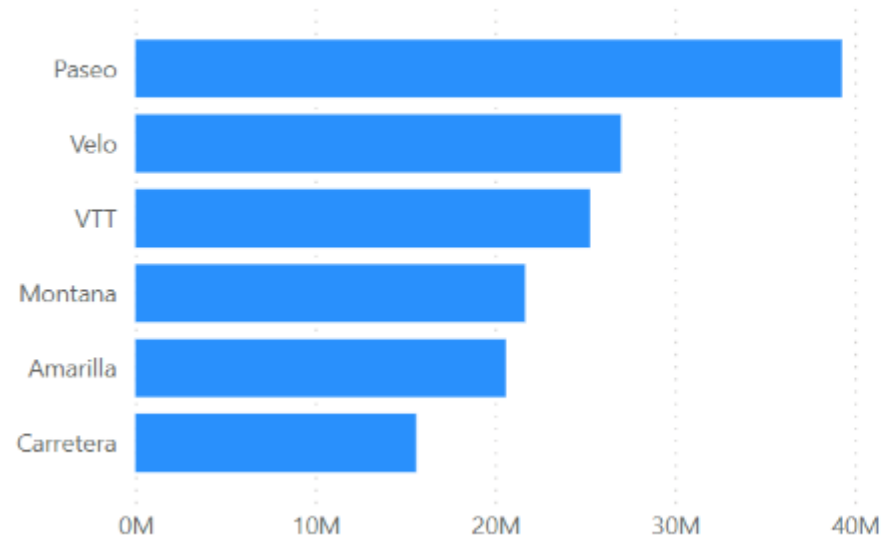
1. Double-click field name in Visualizations Pane
2. Type new name
3. Press Enter

 Use consistent naming across all visuals!

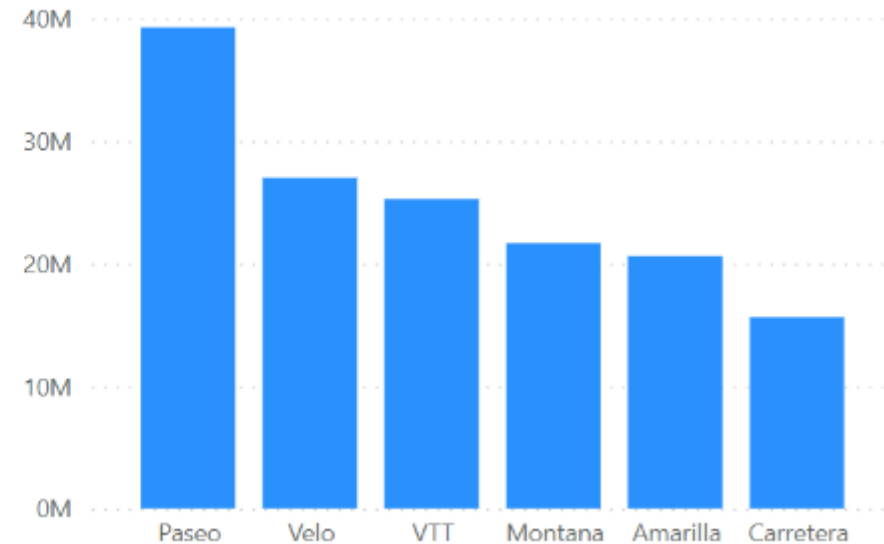
Bar vs. Column Charts

These are the most common charts for comparing data.

Bar Chart



Column Chart



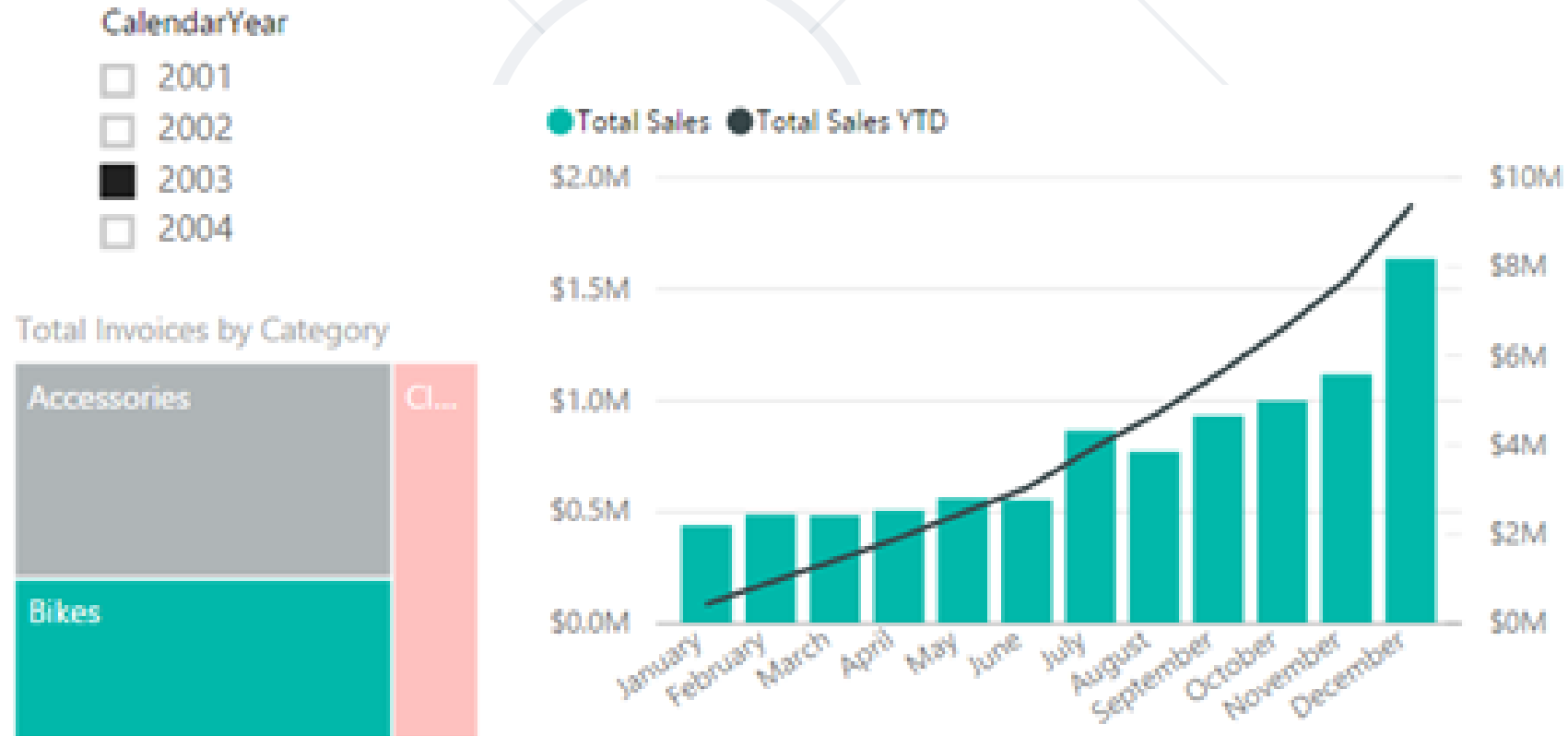
Column Chart (Vertical):

- Use when you have a small number of categories (under 10).
- Best for time periods (Years, Quarters) or short category names.

Bar Chart (Horizontal):

- Best for Long Names (Employees, Products, Cities).
- The human eye scans long text much faster horizontally.

Time Intelligence: The Line Chart



When to use it? Whenever you want to see a Trend or development over time.

Why not a Bar Chart?

- Line Charts connect the data points, showing direction (growth/decline) intuitively.
- Perfect for identifying seasonality and spikes.

Continuous vs. Categorical:

- Power BI automatically detects dates and treats the axis as continuous (allowing you to see the flow of time).

Never use: Bar charts for long-term trends (it looks cluttered).

Table vs. Matrix (Crucial!)

Table Visual (Flat List):

Employee	Month	Sales
Ivan	Jan	1000
Ivan	Feb	1200
Maria	Jan	800
Maria	Feb	900

Use: Detailed lists, audit logs

Matrix Visual (Pivot):

Employee	Jan	Feb
Ivan	1000	1200
Maria	800	900

Use: Cross-tabulations, summaries

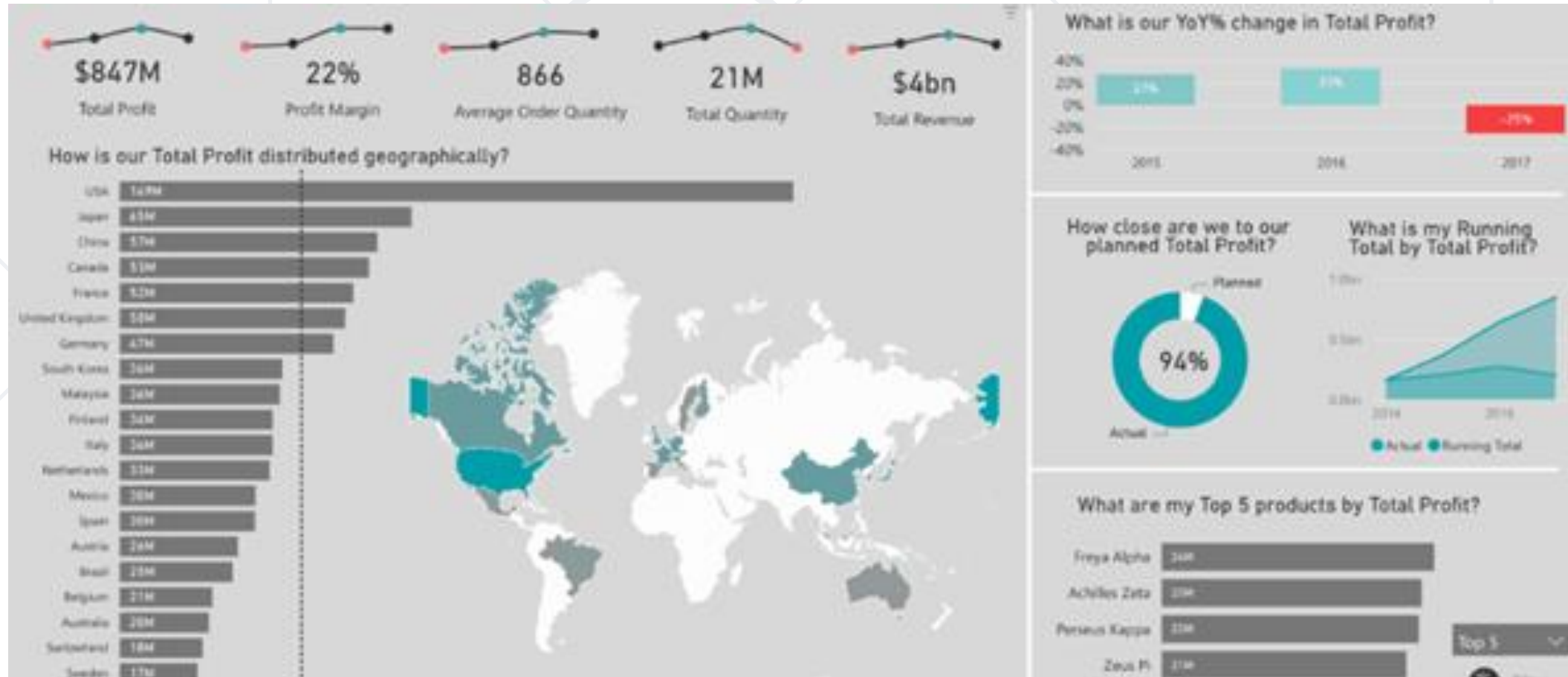
Table Visual:

- A flat list of details.
- Like a standard SQL `SELECT *`.
- Use for: Detailed employee lists, Audit logs.

Matrix Visual:

- A Pivot Table (Rows & Columns).
- Summarizes data.
- Use for: Sales by Year (Columns) and City (Rows).

Geographic Analysis



The Power of Maps: They reveal spatial patterns that spreadsheets hide.

Requirement: Data Categories:

- For maps to be accurate, you must tell Power BI what the text represents.
- Select City column -> **Column Tools** tab -> **Data Category: City.**
- Select Country column -> **Data Category: Country.**

Types:

- **Map (Bubbles):** Bubble size represents the value (Revenue).
- **Filled Map:** Colors the entire region/state based on value.

Pie & Donut Charts

- Rule: Use only if you have 2-4 categories max.
- Good Revenue: Hardware vs Software (2 slices)(filtered category view)
- Bad Revenue by Employee (many slices) → “Confetti Plot”



Comparison Chart

Visual Selection Guide

Visual Type	Best For	Avoid For
Card	Single KPI	Multiple values
Bar (H)	Long names	Time series
Column (V)	Time periods	20+ categories
Line	Trends over time	Comparisons
Table	Detailed lists	Summaries
Matrix	Cross-tabs	Detailed lists
Map	Geography	Non-spatial
Pie	2-3 parts only	5+ categories

Live Demo



Slicers & Filters

- **Slicer** – used on page for interactive functionality allowing users to filter the data and visualizations
- **Filters** – On page filter – pre-defined, normally used to prevent users to select specific data



The 3 Levels of Filters

Visual-Level Filter:

- Affects only one specific chart.
- Ex: "Show Top 5 Employees" on this bar chart only.

Page-Level Filter:

- Affects everything on the current page.
- Ex: "Show data for 2024 only".

Report-Level Filter:

- Affects all pages in the file.
- Ex: "Exclude 'Test' customers from entire report".

- **Scenario:**

We have 100 employees, but the chart is too crowded.

- **Solution:**

Use Top N Filter.



Steps:

1. Select visual
2. Filters Pane → Add filter on "Amount"
3. Filter Type: Top N
4. Show: Top 5
5. By: Sum of Amount

Result:

 Before: 100 bars (unreadable)

 After: 5 bars (clear leaders)

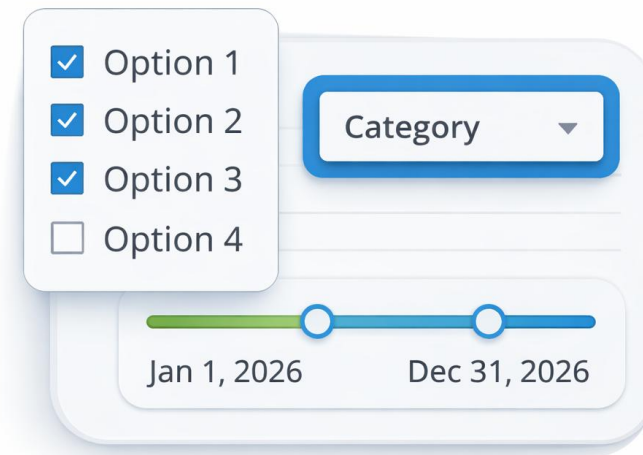
 Tip: Also works for "Bottom N" to see worst performers

Slicers (User Filters)

Definition: On-canvas controls for the end-user.

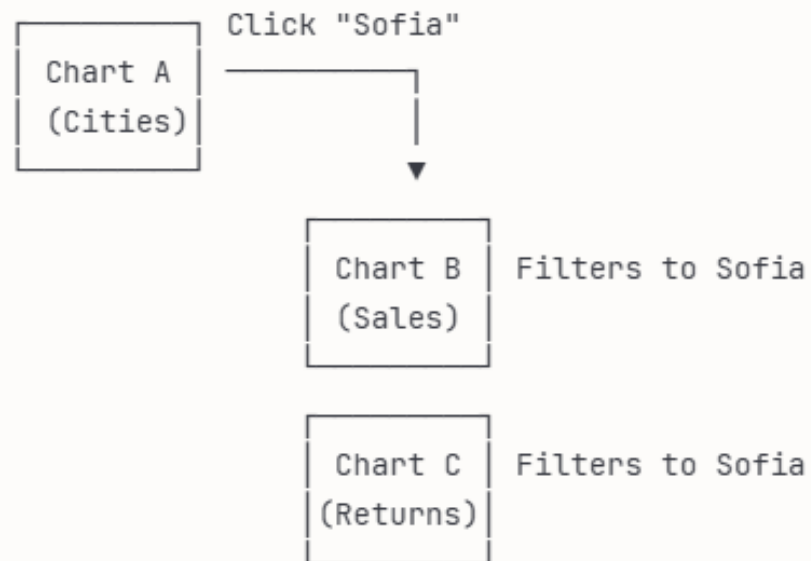
Types:

- **List** (Checkboxes).
- **Dropdown** (Space saver).
- **Range Slider** (Dates).



Default Behavior:

Click bar in Chart A → Filters Chart B, C, D



Edit Interactions:

Format tab → Edit Interactions

- None: Chart doesn't filter
- Filter: Standard behavior
- Highlight: Dims unselected data (keeps all visible)

- **Behavior:** Clicking a bar in one chart filters the rest of the page.
- **Edit Interactions:** You can turn this OFF if you want a visual to remain static (e.g., a "Total Target" card).



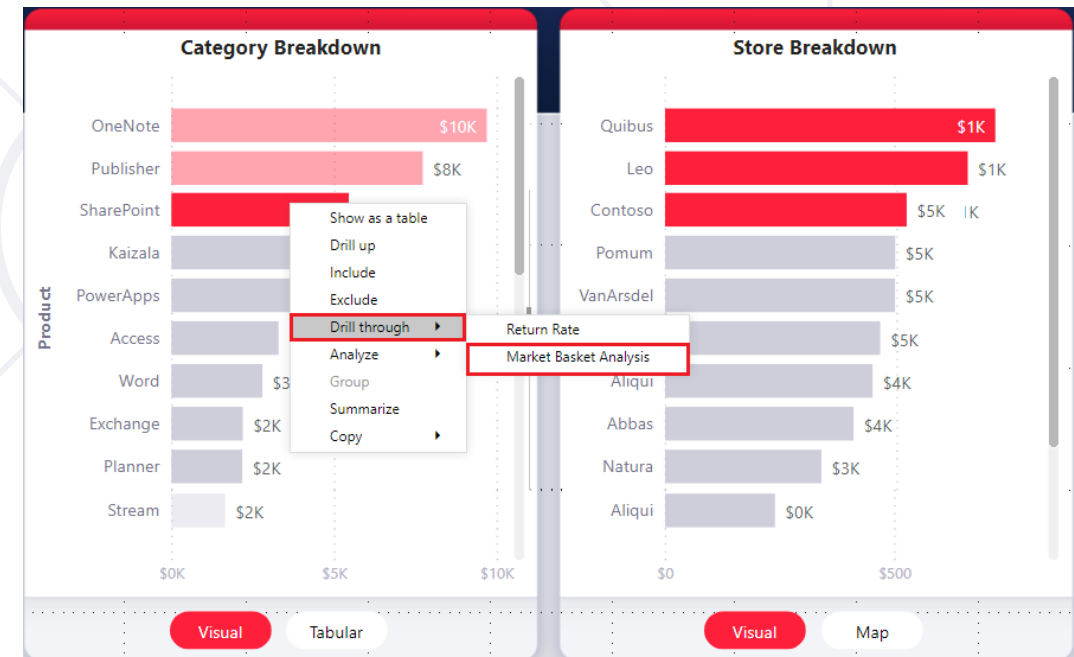
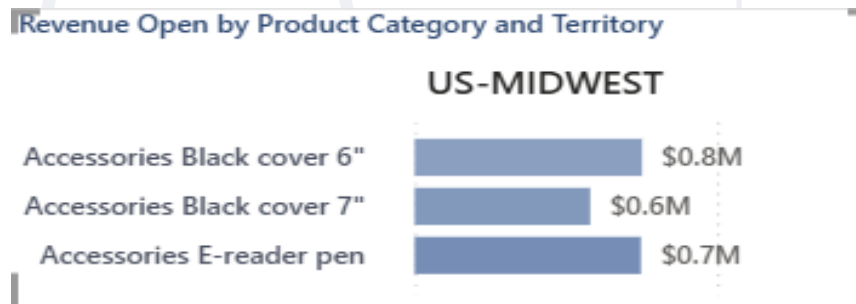
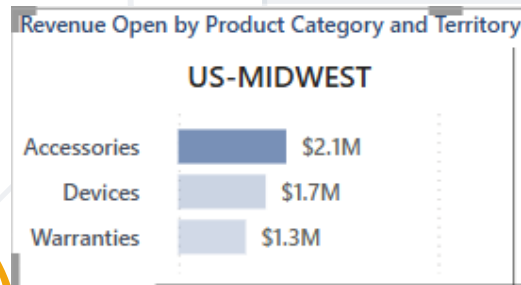
Live Demo



Drill Functionalities

Drill Down – analyzing the details of a graph

Drill Through – allows multi-level report to be connected



- Concept: Grouping data in levels.
- Auto-Hierarchy: Date (Year > Quarter > Month > Day).
- Custom Hierarchy: Country > City > Store.



Drill Down Icons (The Confusion)

- **Drill Mode (Toggle ON/OFF)**

Click this first to enable drilling Then click bar → goes to next level

- **Go to Next Level**

Shows TOTAL for next level Example: 2023 total → Shows Q1+Q2+Q3+Q4 totals

- **Expand All Down (The Fork)**

Shows hierarchy WITH context Example: 2023 → 2023-Q1, 2023-Q2, 2023-Q3...



Most Useful: Expand (🏹) keeps context!

Expand vs. Drill

- Drill Down: You lose the context of the upper level. (You see "January", but which year?).
- Expand: You keep context. (You see "2023 January").
- Recommendation: Usually, "Expand" is more useful for trends.



Drill-Through (Advanced)

- Concept: Right-click a data point -> Jump to a "Details Page".
- Example: Right-click "Sofia" -> Go to "Sofia City Report".

How to **Set Up**:

1. Create detail page (e.g., "City Details")
2. Add field to Drill-through well (City)
3. On summary page: Right-click "Sofia" → "Drill through" → "City Details"
4. Power BI opens detail page filtered to Sofia

Auto "Back" button appears

All visuals on detail page filter to Sofia

Live Demo



Bookmarks & Buttons

- **Bookmarks** – Used for navigations in reports, normally in multi-page report to assign a link to a button
- **Button** – Assign action such as – clear all slicers, Go Back, Go to Bookmark



Purpose: App-like navigation.

Actions:

- Page Navigation (Go to Page 2).
- Web URL (Open Company Site).
- Clear All Slicers (Reset).



Definition: A "Save Game" for your report filters.

Concept: It captures the current state (Slicers, Visibility).

Use Case:

- Button "Show IT Dept" -> Activates a bookmark where IT is filtered.



Bookmarks - Use Cases

Definition: Saves current report state

Common Scenarios:

1. Toggle Views

Bookmark 1: Chart view

Bookmark 2: Table view

Button toggles between them

2. Compare Time Periods

Bookmark 1: Slicers set to 2023

Bookmark 2: Slicers set to 2024

Buttons: "2023" | "2024"

3. Show/Hide Details

Bookmark 1: Summary (details hidden)

Bookmark 2: Expanded (details shown)

Button: "Show More"

What's Saved:

Slicer selections

Visual visibility

Filter states

Current page

- **Standard Tooltip:** Hover over a bar -> See the number.
- **Report Page Tooltip:** Hover over a bar -> See a mini-chart inside the popup.



Live Demo



Conditional Formatting

Conditional formatting allows for easy grasping and identifying outliers, targets, etc., based on logic, by rule or another value. It can color the value, add icon, etc. KPI, color grading. It is applied to tables or matrix

Background color – Text Color



Background color - Affordability rank

Format style: **Gradient** (selected)

Apply to: Values only

What field should we base this on?: Affordability rank

Summarization: Sum

How should we format empty values?: As zero

Minimum: Lowest value (selected)

Center: Middle value (selected)

Maximum: Highest value (selected)

☒ Add a middle color

[Learn more about conditional formatting](#)

OK Cancel

Top 10 Weather States with Affordability and Overall Rank			
State	Weather rank	Affordability rank	Overall rank
Arizona	1	16	6
Kentucky	2	14	9
West Virginia	3	21	11
Georgia	4	3	1
California	4	48	37
Massachusetts	6	42	10
New Mexico	7	10	19
Tennessee	8	1	3
Arkansas	9	19	44
Hawaii	9	50	33

Why Color Matters?

- Preattentive Processing: The brain notices color before reading numbers.
- Goal: Highlight "Good" vs "Bad".



Conditional Formatting

Background color – Background color



Background color - Affordability rank

Format style: Rules Apply to: Values only

What field should we base this on? Affordability rank

Summarization: Sum

Rules

If value	Operator	Value	Unit	and	Operator	Value	Unit	then	Color
>=		0	Percent	and	<	33	Number	then	Green
>=		34	Percent	and	<	66	Number	then	Yellow
>=		67	Percent	and	<	100	Number	then	Red

[Learn more about conditional formatting](#)

OK Cancel

Top 10 Weather States with Affordability and Overall Rank

State	Weather rank	Affordability rank	Overall rank
Arizona	1	16	6
Kentucky	2	14	9
West Virginia	3	21	11
Georgia	4	3	1
California	4	48	37
Massachusetts	6	42	10
New Mexico	7	10	19
Tennessee	8	1	3
Arkansas	9	19	44
Hawaii	9	50	33

- **Background:** Great for Heatmaps in Matrices (Darker = Higher Sales).
- **Font:** Great for KPIs (Red text if negative).

Background color - % revenue region

Format style

Rules

Apply to

Values only

What field should we base this on?

Sum of % revenue region

Summarization

Sum

Rules

Reverse color order

New rule

If value

>=

0

Number

and

<

0.25

Number

then

↑

↓

×

If value

>=

0.26

Percent

and

<

0.41

Number

then

↑

↓

×

If value

>=

0.42

Percent

and

<

1

Number

then

↑

↓

×

Learn more about conditional formatting

OK

Cancel

Region	Revenue	% revenue region
West	1,129,702,160.48	21.73%
Central	1,483,669,485.77	28.53%
	2,104,731,284.53	40.48%
East	2,306,402,738.65	44.36%
Total	5,199,504,107.30	100.00%

Conditional Formatting

Background color – Data Bars



Data bars - Affordability rank

Format style: Data bars (dropdown)
Apply to: Values only (dropdown)

What field should we base this on?: Affordability rank1 (dropdown)
Summarization: Sum (dropdown)
Bar direction: Left to right (dropdown)

Minimum: Lowest value (dropdown)
Enter a value:
Maximum: Highest value (dropdown)
Enter a value:

Positive bar: ☒ (dropdown)
Negative bar: ☐ (dropdown)
Axis: ☒ (dropdown)

☐ Show bar only

[Learn more about conditional formatting](#)

OK Cancel

TABLE: Top 10 weather states with afford   ...

State	Weather	Affordability	Overall rank
Hawaii	1	45	10
Florida	2	25	5
Louisiana	3	29	36
Texas	4	24	17
Georgia	5	19	28
Mississippi	6	6	19
Alabama	7	10	16
South Carolina	8	27	41
Arkansas	9	4	11
Arizona	10	33	38

- What: Mini bar charts inside a table cell.
- Why: Allows you to scan a list of 50 employees and instantly see who is top performing without reading 50 numbers.



Conditional Formatting

Background color - Icons

Icons - % Change vs Prior Year

Format by **Rules** [Learn more](#)

Based on field **% Change vs Prior Year**

Icon layout: Left of data
Icon alignment: Top

Rules

Condition	Value	Unit	Operator	Value	Unit	Icon
If value is greater than or equal to	0	Percent	and is less than	33	Percent	Red Diamond
If value is greater than or equal to	33	Percent	and is less than	67	Percent	Yellow Triangle
If value is greater than or equal to	67	Percent	and is less than or equal to	100	Percent	Green Circle

OK Cancel

TABLE: Top 10 weather states with afford   ...

State	Affordability	Weather	Overall rank
Hawaii		45	1
Florida		25	2
Louisiana		29	3
Texas		24	4
Georgia		19	5
Mississippi		6	6
Alabama		10	7
South Carolina		27	8
Arkansas		4	9
Arizona		33	10



- **Traffic Lights:**   
- **Arrows:**  
- **Logic (Rules):**
 - If Sales < Target -> Red Flag.
 - If Sales > Target -> Green Check.



Live Demo



What is DAX?

- Data Analysis Expressions.
- It is the formula language of Power BI.
- Analogy: It's like Excel formulas, but it works on entire columns, not cells.



When to use **CALCULATED COLUMN**:

Row-by-row calculation

Need to filter/slice by result

Example: Category = IF(Amount > 1000, "High", "Low")

When to use **MEASURE**:

Aggregation (SUM, COUNT, AVG)

Dynamic calculation (changes with filters)

Example: Total Sales = SUM(Sales[Amount])

Do you need to SUM/COUNT/AVG?

| YES → Use MEASURE 

| NO ↓



Do you need to filter by it?

| YES → Use CALCULATED COLUMN

| NO → Use MEASURE anyway (smaller file)

Golden Rule: Default to MEASURES!



Why use Measures?

- If you drag a column to a visual, Power BI creates an "Implicit Measure".
- But writing your own "Explicit Measures" is safer and reusable.
- Rule: Always use Measures for aggregations (Sums, Counts, Averages).



- Goal: Calculate Total Revenue
- Formula: *Table name depends on the data source (e.g. vw_fct_sales)

Total Revenue = $\text{SUM}(\text{'vw_fct_sales' [Amount]})$

Total Cost = $\text{SUM}(\text{'vw_fct_sales' [Total Cost]})$

Total Profit = $[\text{Total Revenue}] - [\text{Total Cost}]$

How: Right-click table -> New Measure.



Live Demo



The 10 Commandments of Design

1.  1. Less is More: Max 5-7 visuals per page
2.  2. Z-Layout: KPIs top-left (eye starts here)
3.  3. Consistent Colors: Same metric = same color
4.  4. Clear Titles: "Sales by Employee" not "Sum of Amount by Name"
5.  5. Format Numbers: 1.2M not 1,234,567.89
6.  6. Avoid Pie Charts: Use only for 2-3 slices max
7.  7. No 3D Charts: They distort data perception
8.  8. Don't Cram: White space is good!
9.  9. Use Tooltips: Hide details, show on hover
10.  10. Test Mobile View: 50% of users are on phones



- Avoid Red/Green combinations without icons.
- Use high contrast.
- Use "Color Blind Safe" themes (View tab -> Themes).



- Power BI reports are landscape (16:9).
- Phones are portrait.
- Feature: Mobile Layout View allows you to rearrange visuals specifically for phone users.



- Complete interface understanding
- All fundamental visuals
- Filter hierarchy mastery
- Drill functionalities
- Interactive features (buttons/bookmarks)
- DAX basics
- Professional design principles



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